

Key Management Techniques

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A Unifying Solution – So Far!

Use of shared secret keys at the legitimate entities

Construct problems

1. Easy to solve with key
2. Hard to solve without key

Bounded computational complexity at the adversary

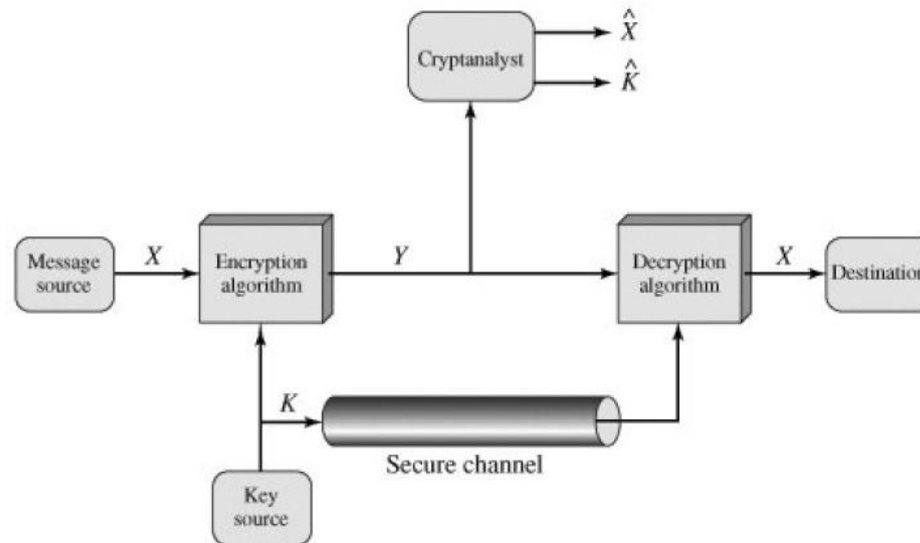
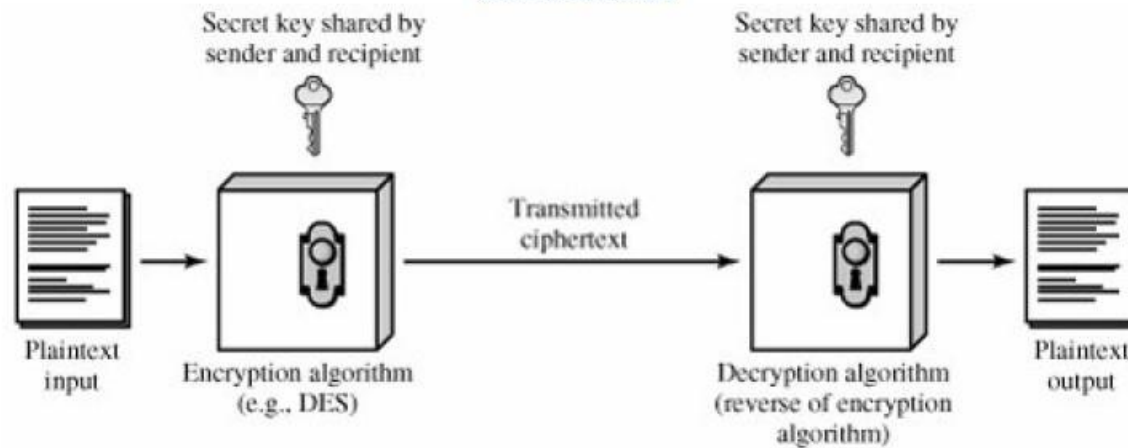


*google images

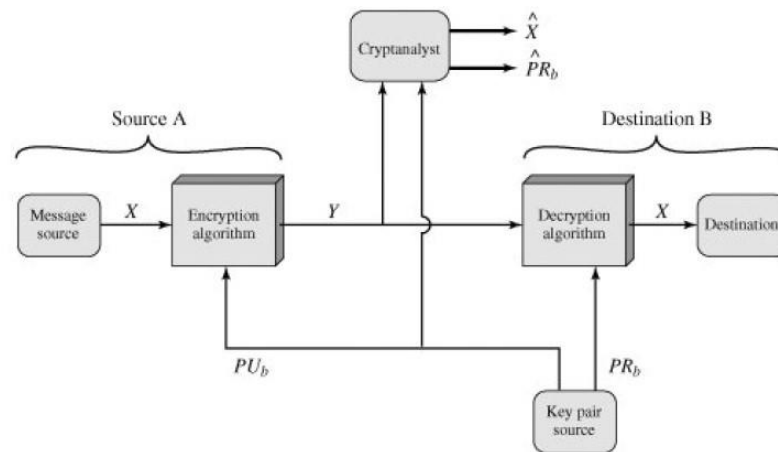
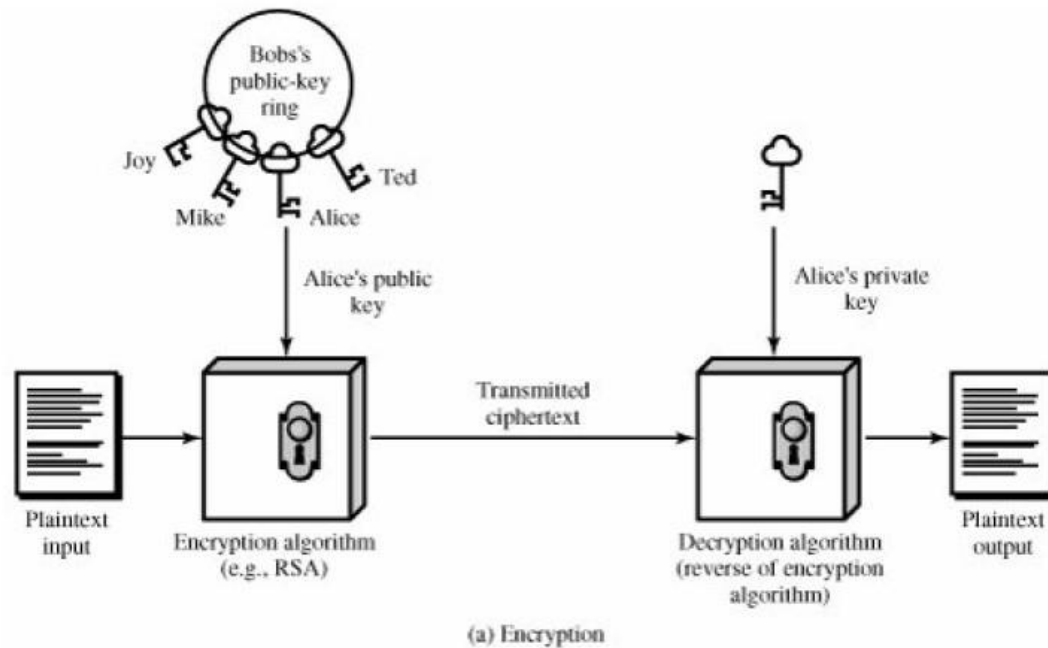
Outline

1. Motivation for key management
 1. Symmetric key encryption
 2. Asymmetric key encryption
2. Key distribution problem
 1. Key distribution center
 2. Key hierarchy
 3. Decentralized control
3. Distribution of public keys
 1. Use of public key cryptography for key exchange
4. New developments in key generation

Symmetric Key Encryption



Asymmetric Key Encryption - Confidentiality



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Key Distribution Problem

1.

A can select a key and physically deliver it to B.

2.

A third party can select the key and physically deliver it to A and B.

3.

If A and B have previously and recently used a key, one party can transmit the new key to the other, encrypted using the old key.

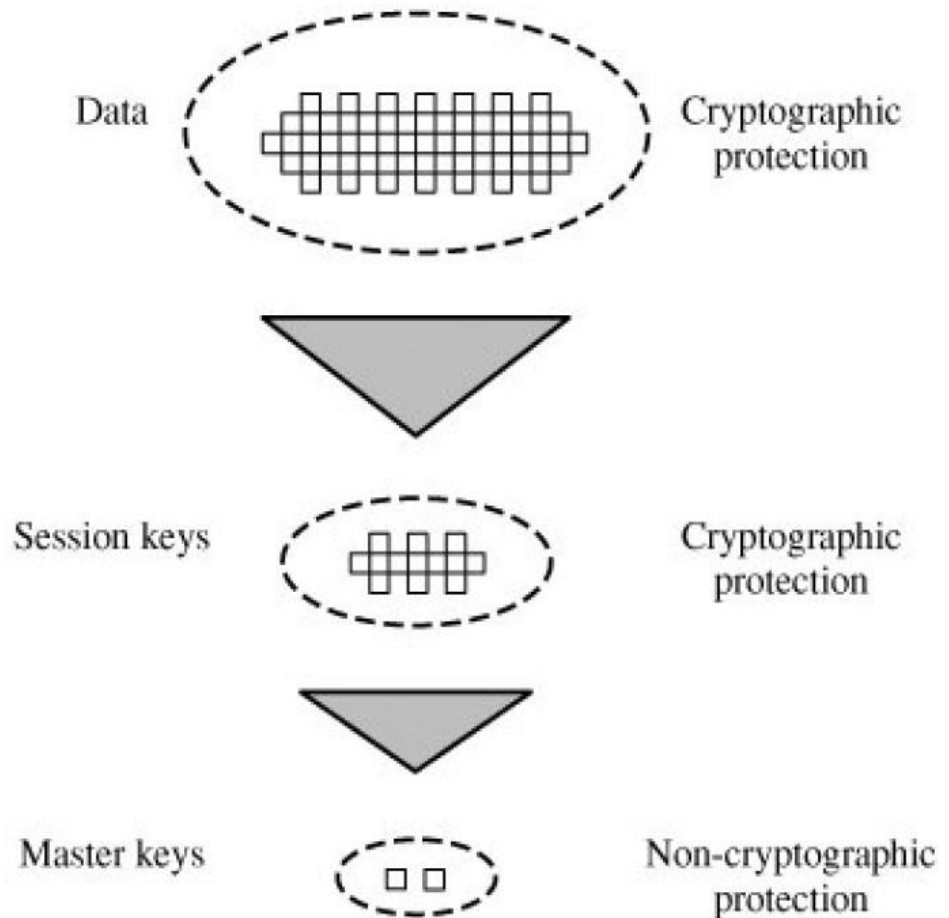
4.

If A and B each has an encrypted connection to a third party C, C can deliver a key on the encrypted links to A and B.

Key Distribution Problem

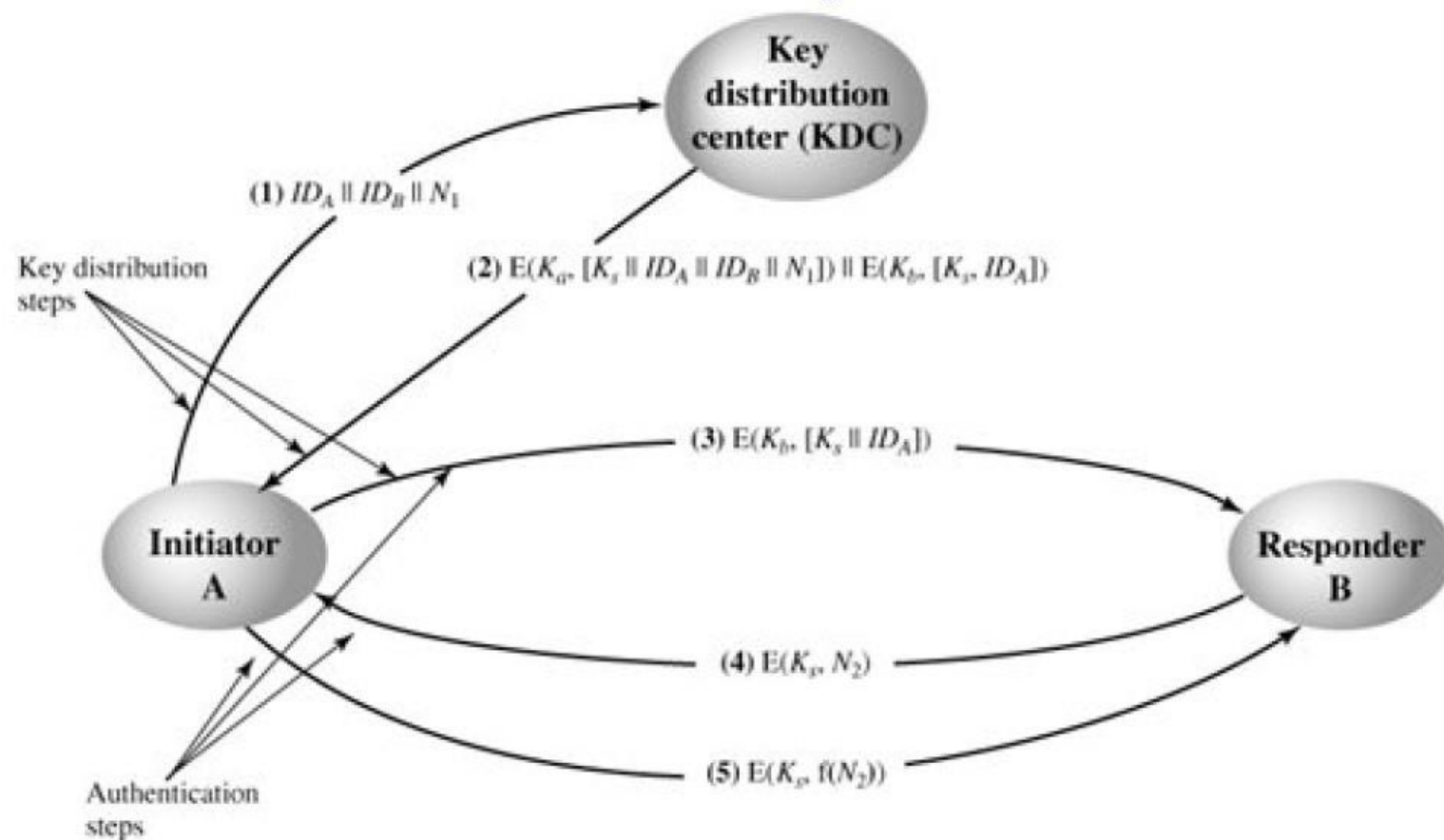
Key Hierarchy

1. Session keys are for the nodes
2. Session keys for short duration
3. Master key for the distribution center
4. Master keys are used to generate session keys



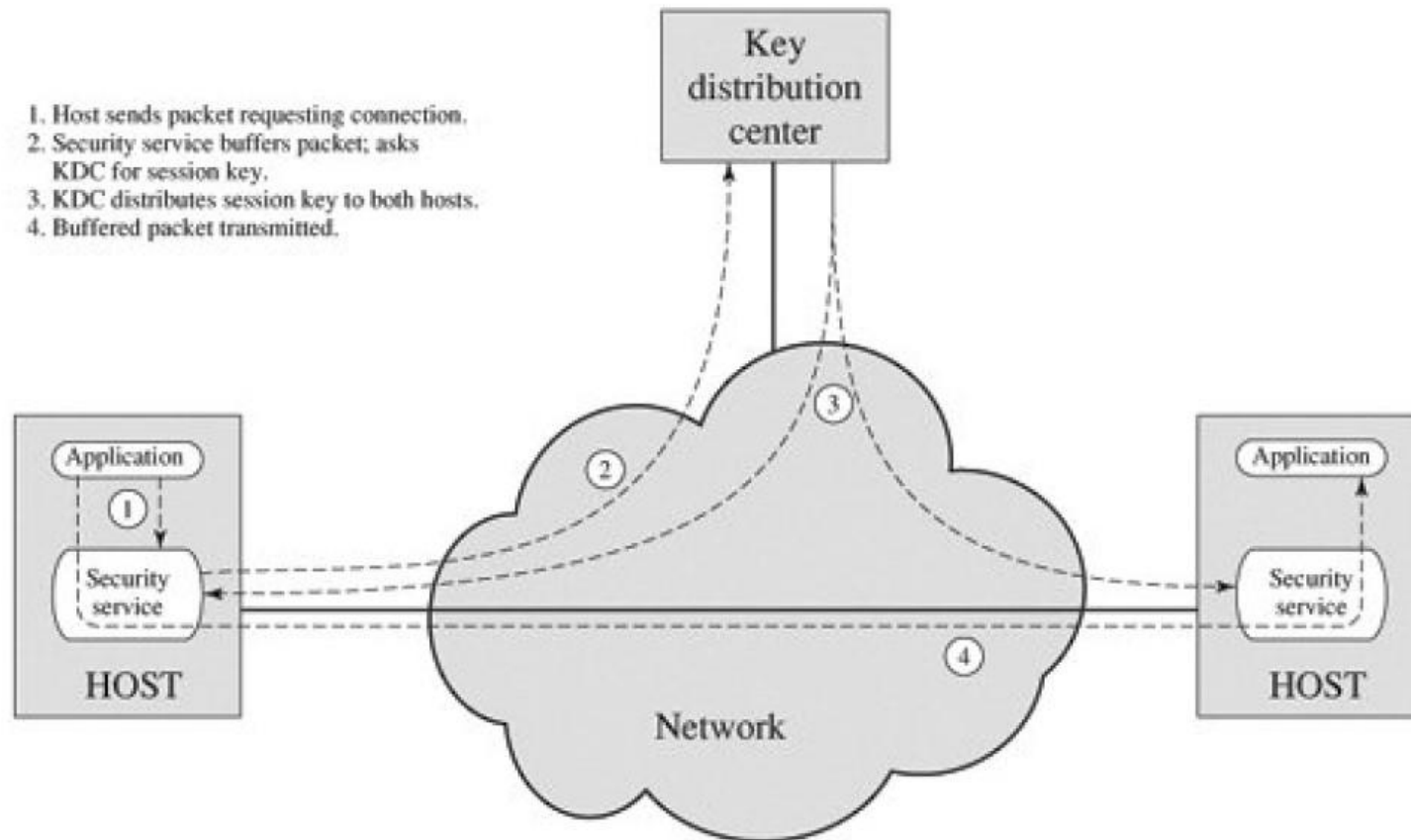
Key Distribution Problem - Centralized

Transaction with KDC – Master key for session keys



Key Distribution Problem - Centralized

How do several applications use this feature?

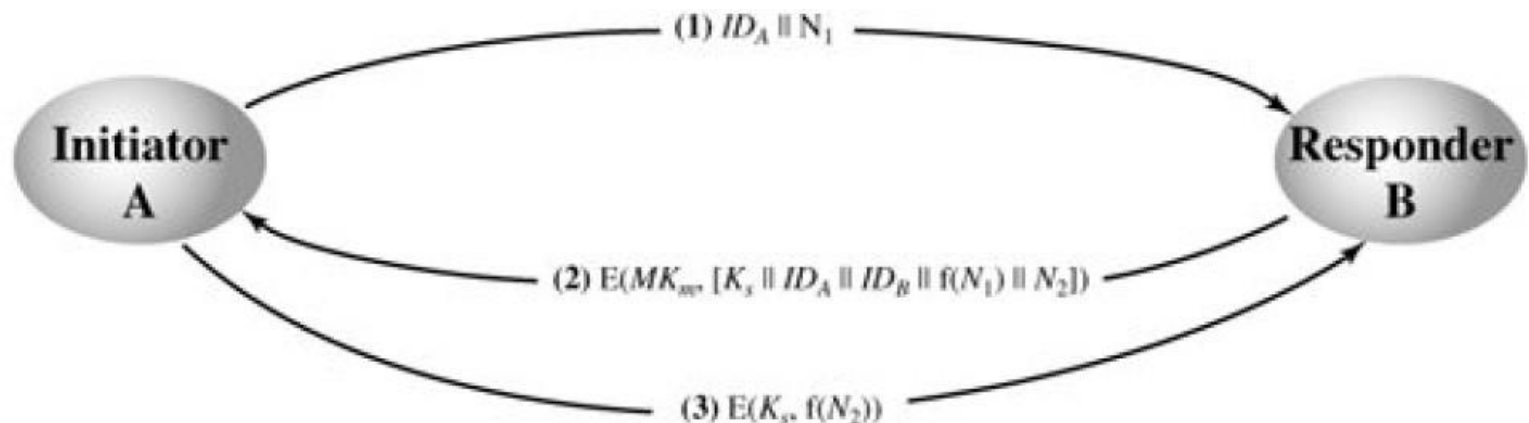


Key Distribution Problem - Decentralized

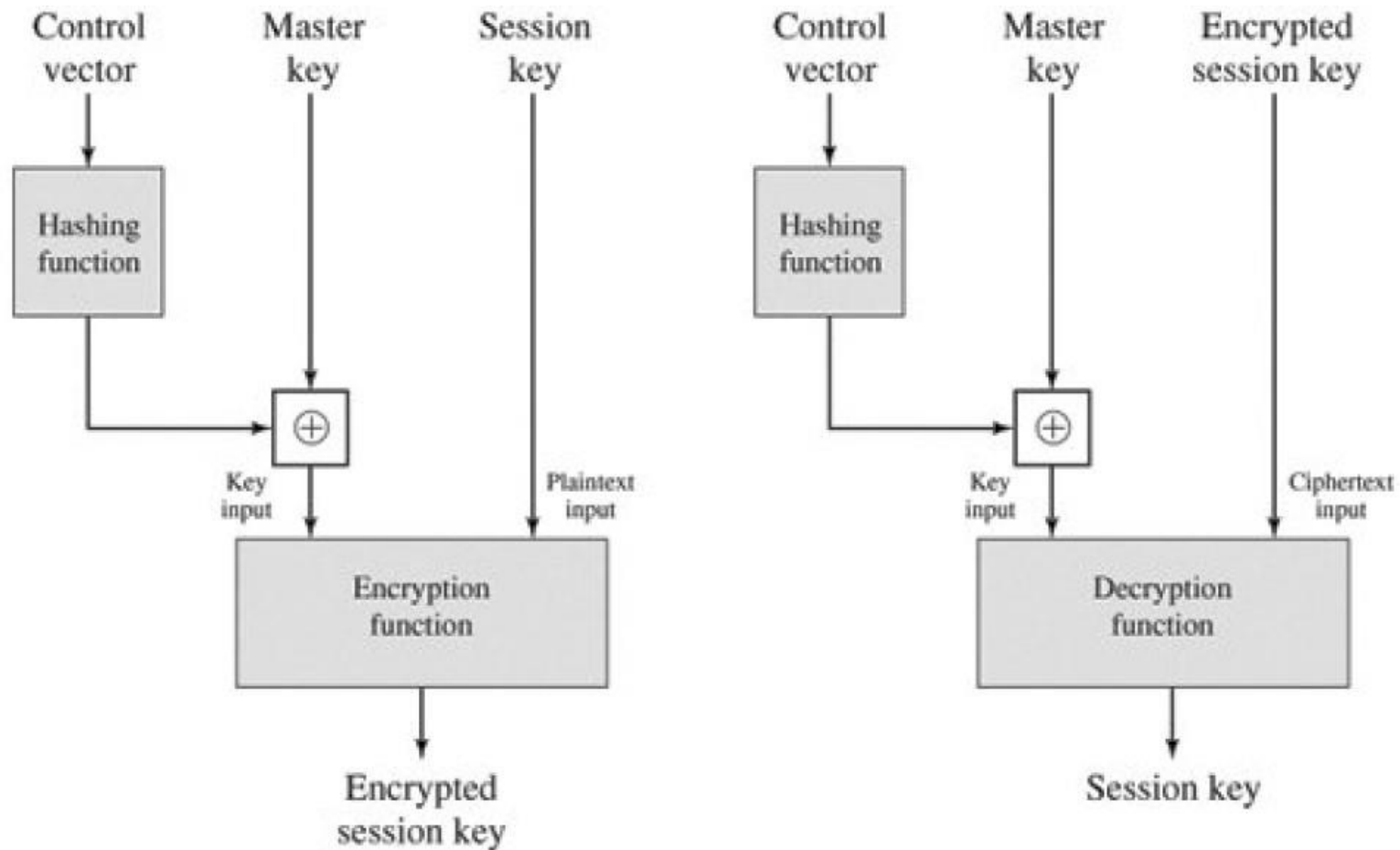
KDC must be trusted and protected

Decentralized ideas can be helpful in some contexts

Need more master keys compared with the scheme using KDC



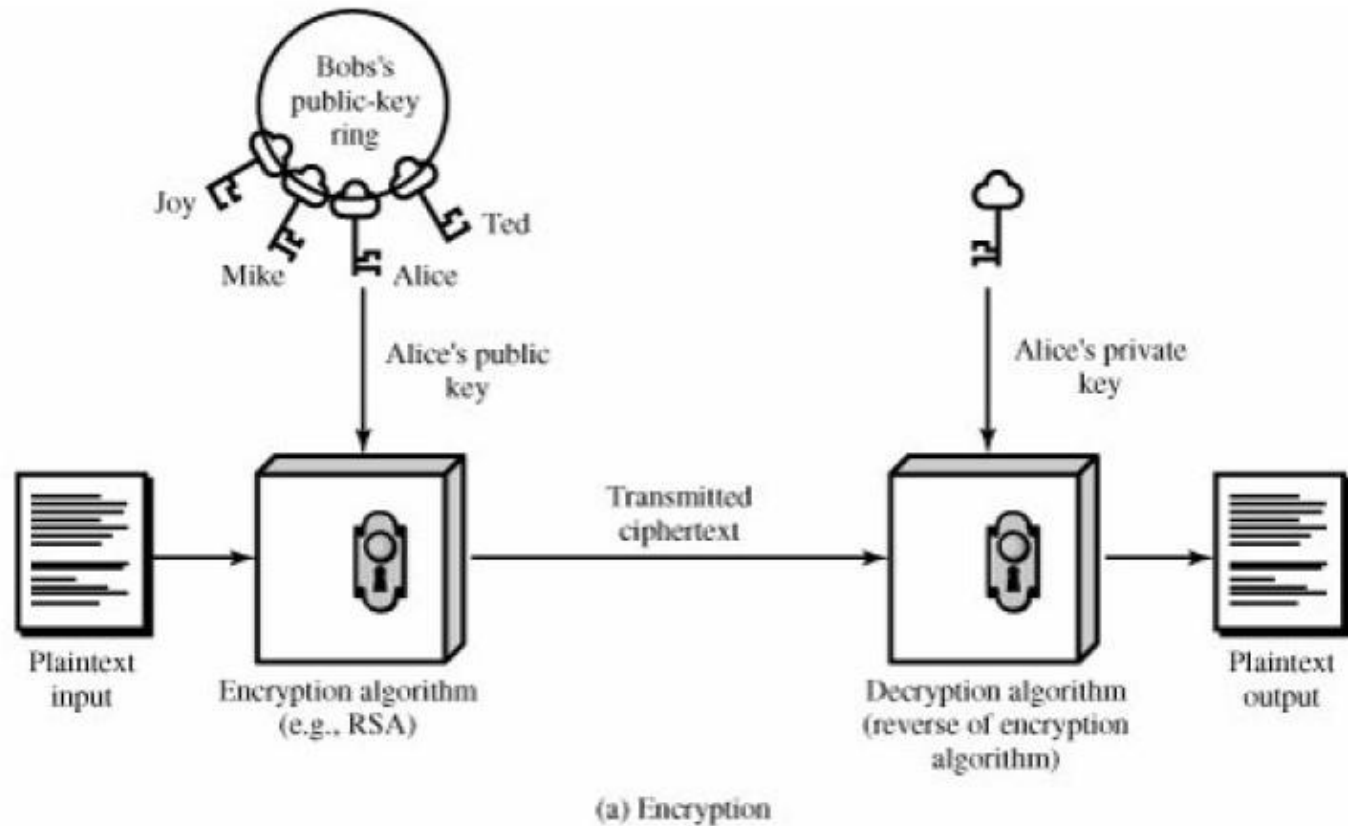
Key Distribution Problem – Controlling Key Usage



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 1. Use of public key cryptography for key exchange
4. Some basics on discrete logarithm
5. Diffie-Hellman key exchange
 1. Algorithm and protocols
6. New developments in key generation

Recall Public Key Cryptography



There are two keys – public key and private key

Public key must be shared with the users that wish to securely communicate

Distribution of Public Keys

Objectives:

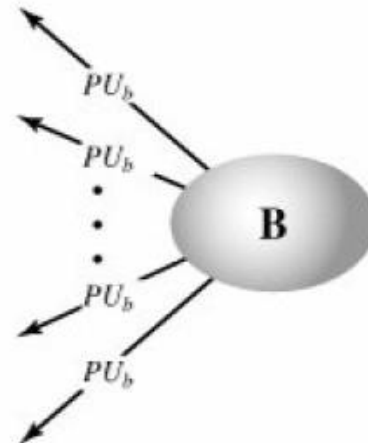
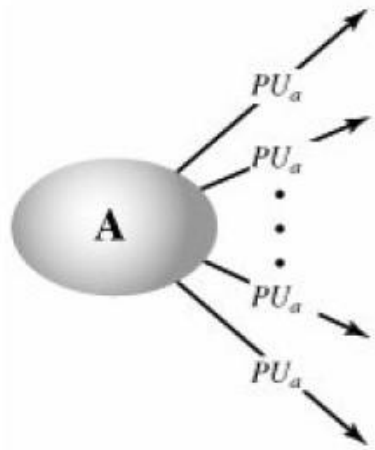
1. Distribution of public keys to the users – for secure communication and authentication
2. Use of public key encryption to distribute secret keys

Distribution of public keys

1. Public announcement
2. Public directory
3. Public key authority
4. Public key certificates

Distribution of Public Keys

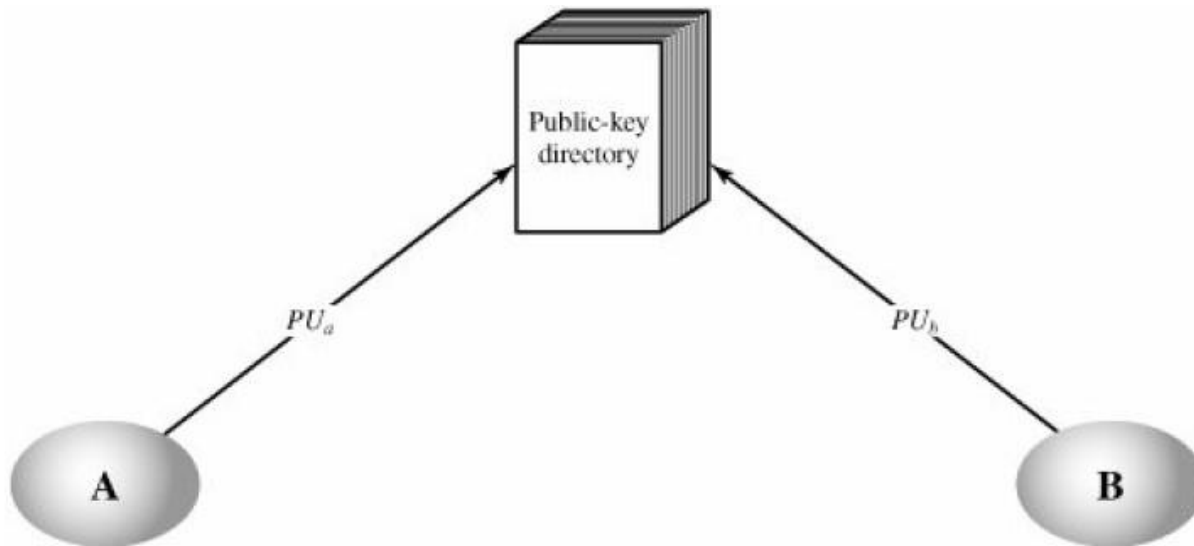
Public Announcement



Vulnerable to impersonation attack

Distribution of Public Keys

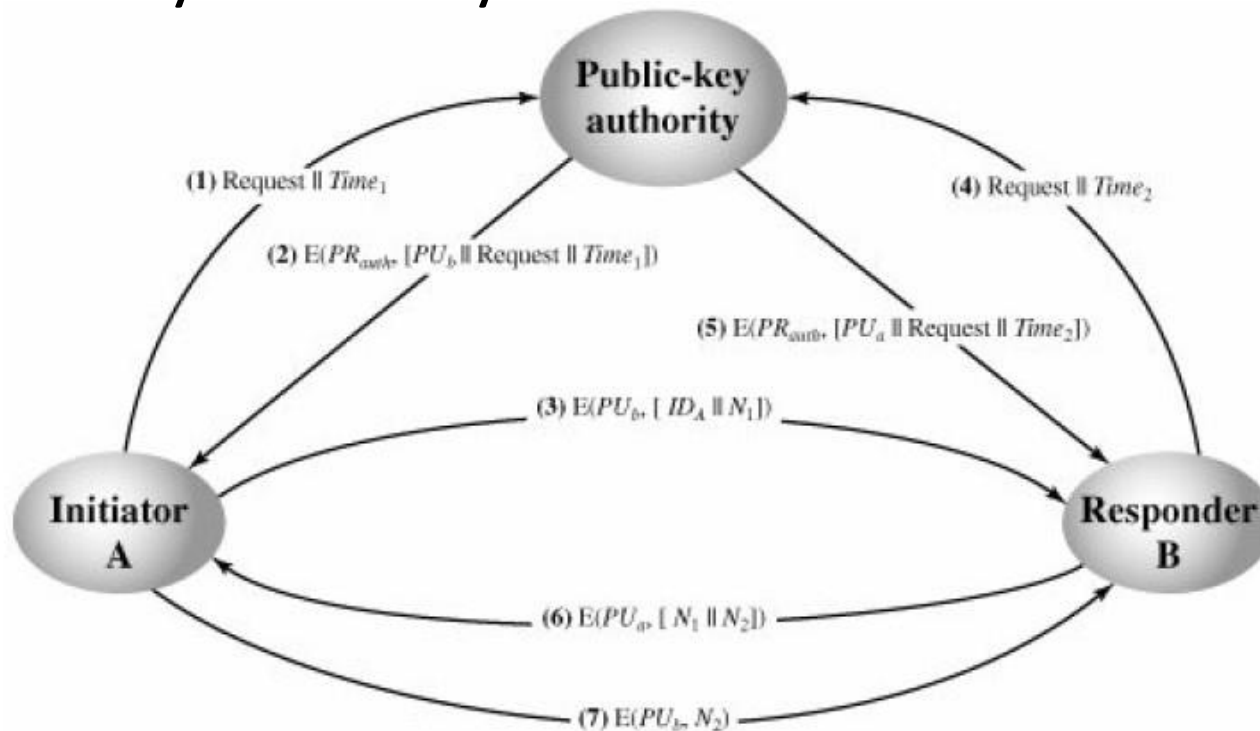
Public Key Directory



Vulnerable to threats if the directory is compromised

Distribution of Public Keys

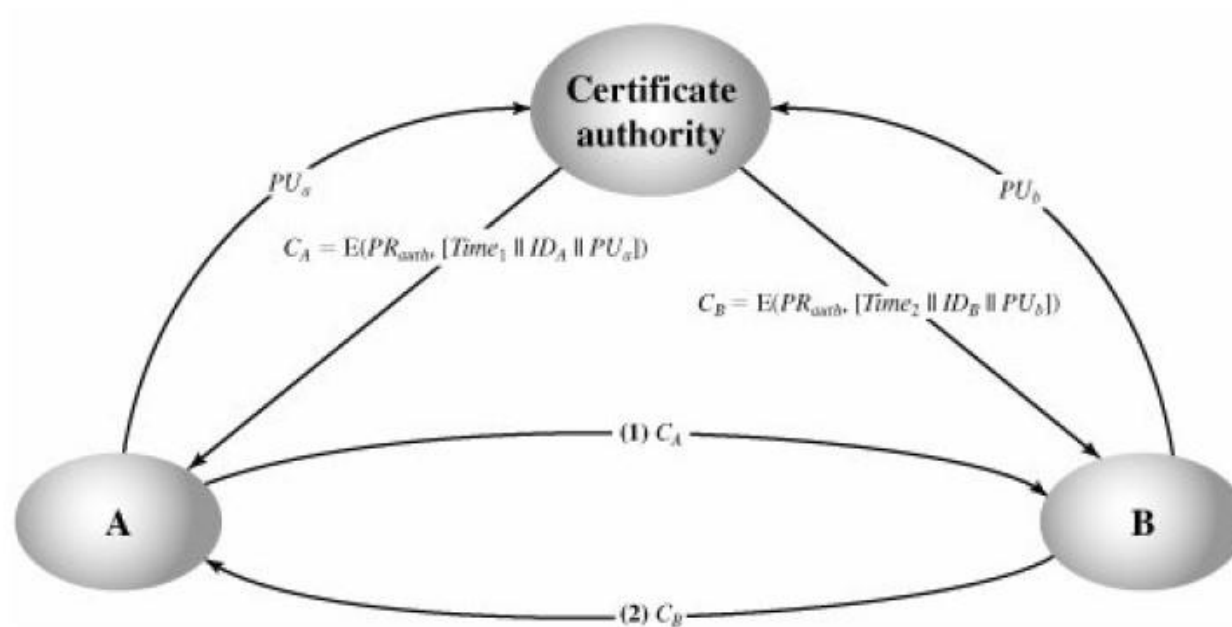
Public Key Authority



Users can authenticate the authority using its public key

Distribution of Public Keys

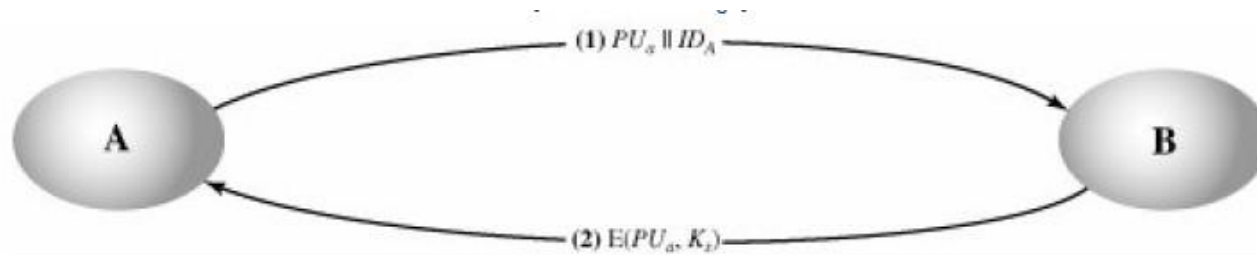
Public Key Certificates



Users pass the certificate that is signed by the authority

Distribution of Public Keys

Public Key distribution to share secret keys



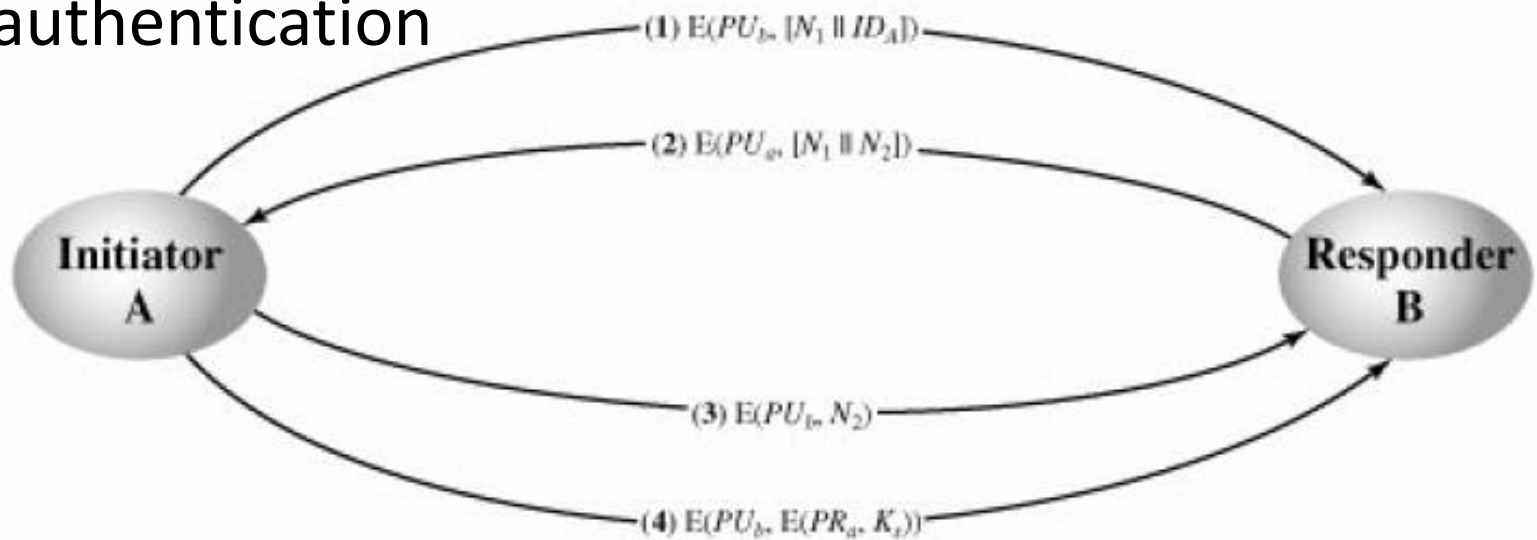
A shares the public key to B

B generates a secret key and then encrypts it using the public key

*However this is again vulnerable impersonation attack

Distribution of Public Keys

Public Key distribution to share secret keys with authentication



A and B already have each other's public key

Presence of N_1 in (2) confirms that only B opened the message

Presence of N_2 in (3) authenticates A

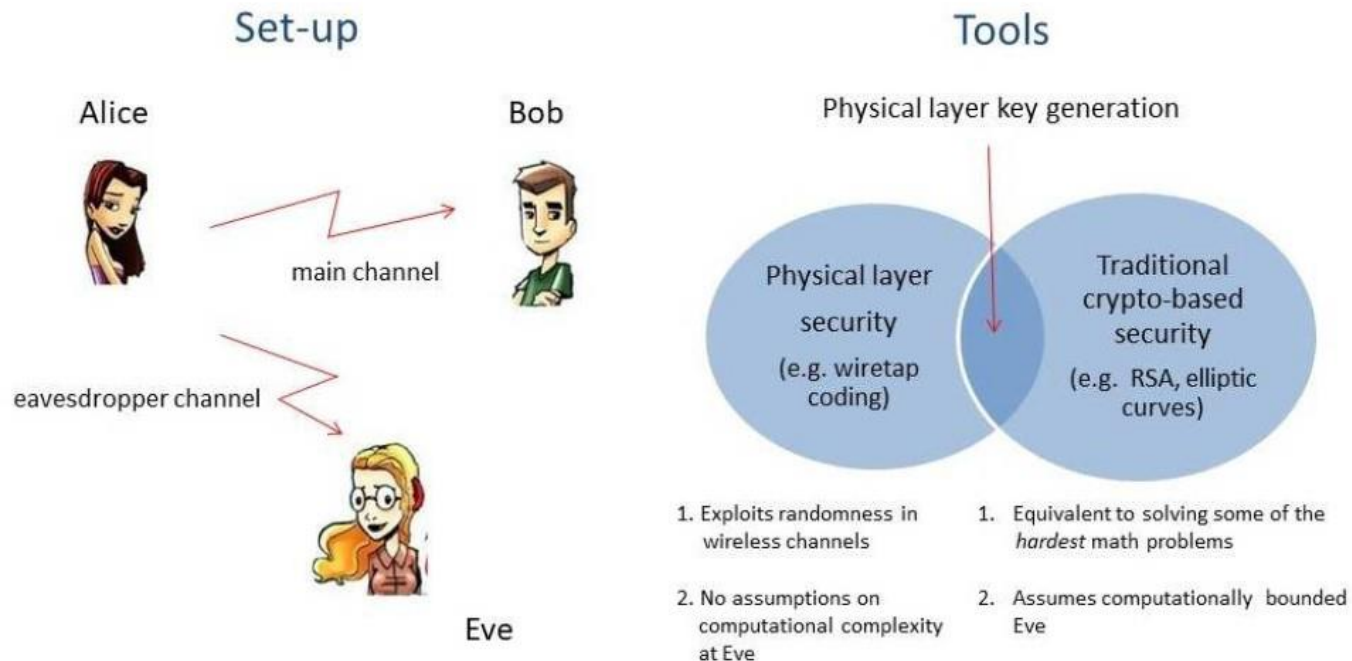
Finally A sends the secret key so that only B gets it and B knows it was sent by A

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New Developments – Dynamic Key Generation

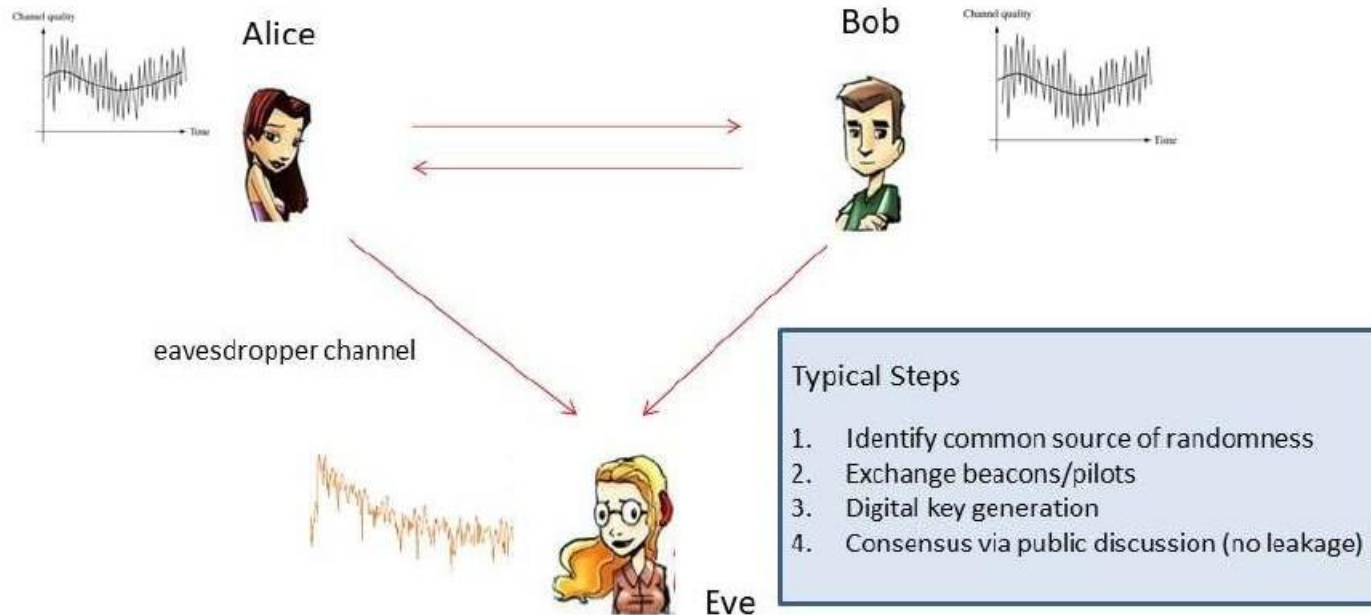
Physical layer security vs. traditional crypto-based security



New Developments - Dynamic Key Generation

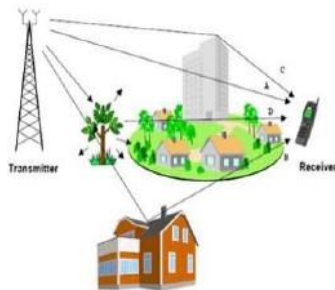
Physical Layer Key Generation – Modelling assumptions

Physical layer key generation

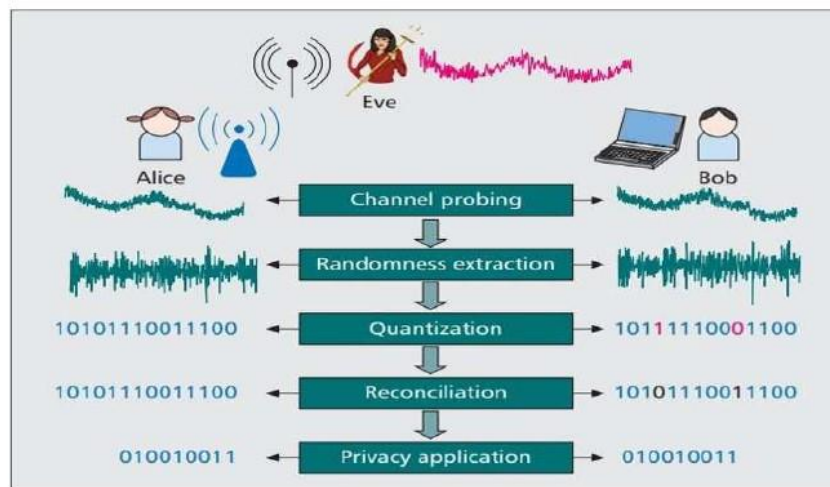
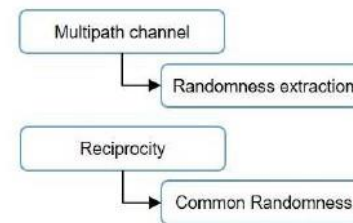


New Developments - Dynamic Key Generation

Principles of Key Generation



Channel Characteristics



New Developments - Dynamic Key Generation

Recent research contributions



Physical layer secret key generation for fiber-optical networks

Konstantin Kravtsov,^{1,*} Zhenxing Wang,² Wade Trappe,³ and Paul R. Prucnal⁴

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²Finisar Corporation, 200 Precision Road, Horsesham PA, 19044 USA

³WINLAB, Rutgers University, North Brunswick NJ, 08902 USA

⁴Department of Electrical Engineering, Princeton University, Princeton NJ, 08544 USA

Radio-telepathy: Extracting a Secret Key from an Unauthenticated Wireless Channel

Suhas Mathur, Wade Trappe, Narayan Mandayam, Chunxuan Ye¹, Alex Reznik²

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Secret key generation through OFDM multipath channel

3

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Abstract

Authors

References

Cited By

Keywords

Metrics

Physical Layer Wireless Security Made Fast and Channel Independent

Shyamnath Gollakota and Dina Katabi
Massachusetts Institute of Technology

The Effect of Eavesdropper's Statistics in Experimental Wireless Secret-Key Generation

Alexandre J. Pierrot*, Student Member, IEEE, Rémi A. Chou, Student Member, IEEE and Matthieu R. Bloch, Member, IEEE

Summary

Use of shared secret keys at the legitimate entities

Construct problems

1. Easy to solve with key
2. Hard to solve without key

Bounded computational complexity at the adversary

Overheads?

Latency of crypto-solution?

Large-scale networks?

Post-quantum?



*google images

Questions?

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References:

Cryptography and Network Security, Principles and Practices, by William Stallings
(combination of several editions)

Cryptography and network security course at SCSE, NTU by Prof. Anwitaman Datta